

Shadow-Decoy + global_max: A Two-Round Scoring Variant

N. Wamsley

2026-07-17

Motivation

Pioneer’s single-pass LGBM (round 1) scores each PSM using per-run features. On its own it has no way to say “the same peptide was also confidently identified in another run.” A two-round scheme adds a **cross-run consistency feature** and re-trains a second LGBM.

Two design choices control how aggressive that feature is, and whether the LGBM is allowed to lean on it as a standalone signal:

1. **What is the cross-run feature?** — a summary of the same precursor’s round-1 scores in *other* runs.
2. **How much can the LGBM trust it?** — do we regularise so the marginal on the feature alone can’t discriminate, forcing joint use with the base features?

shadow + global_max is one specific answer to both.

Pipeline

```
round-1 LGBM (per-PSM features)    -> per-run s1 score (00F via CV folds)
  |
  v
write cross-run column: global_max  -> mbr_score per (precursor, run)
  |
  v
inject symmetric shadow rows        -> double the row count per fold-file
  |
  v
round-2 LGBM (all features + mbr_score)
  |
  v
compute FDR-A (real-only) and FDR-B (shadow-inclusive)
  |
  v
remove shadows                      -> downstream sees the same schema
```

Both new steps sit between the round-1 and round-2 LGBM invocations in `_score_precursor_isotope_traces_no_mbr` (see `two_round_scoring.jl`). They only fire when the environment variable `TWO_ROUND=1` is set AND `match_between_runs` is false in config.

The global_max feature

Let $s1(p, r)$ = round-1 out-of-fold LGBM score for precursor p in run r . For each PSM at (p, r) , define:

$$\text{global_max}(p, r) = \max_{r' \neq r} s1(p, r')$$

with 0 when p doesn't appear in any other run. This is written as a new column into every per-file fold Arrow. It's a strong "does this precursor appear anywhere else in the experiment?" signal: **one** confident hit in any other run yields a high value.

Two design contrasts against the earlier knn5_mean:

aspect	knn5_mean	global_max
scope	top-5 most similar runs (cosine on s1 profile)	every other run
aggregation	mean	max
leakage risk	low if top-5 are all same-condition	higher — 1 cross-condition run is enough

Shadow-decoy injection

After `mbr_score` (= `global_max` here) is written, but **before** round-2 training, we inject shadow rows symmetrically.

Per fold-file:

- For every real target T : create a **shadow_decoy** row = nearest-iRT decoy's columns, with `mbr_score` **overwritten** by T 's value. Labeled decoy.
- For every real decoy D : create a **shadow_target** row = nearest-iRT target's columns, with `mbr_score` overwritten by D 's value. Labeled target.

An `is_shadow` boolean column flags the injected rows (false on originals). Row count exactly doubles when both classes are present in a file.

Each shadow inherits its **source row's** `precursor_idx` and `cv_fold`, so the precursor-keyed CV invariant (all instances of a precursor share one fold — required for the round-1 scores to be OOF) remains intact.

Why symmetric

At every `mbr_score` value that came from a target T , we now have:

- 1 target-labeled row (the real T)
- 1 decoy-labeled row (T 's shadow_decoy)

and analogously for every value from a real decoy. **The marginal on `mbr_score` alone is uninformative** — the target/decoy ratio at each `mbr_score` value is exactly 1:1.

The round-2 LGBM is therefore forced to use `mbr_score` **jointly** with the base features: "high `mbr_score` AND target-like base features $\rightarrow$ target". A high `mbr_score` on a row with decoy-like base features (as in the shadow_decoy) doesn't get promoted.

Empirically, the LGBM’s total split-gain on `mbr_score` drops from ~**86%** (no shadows) to ~**13%** (with shadows) on the FTR-27 benchmark. The base features re-emerge as the dominant signal.

Round-2 training and FDR

The round-2 trainer is unchanged — it just sees a fold-file whose row count is roughly $2\times$ what it was, with the correct target/decoy labels on every row. The `is_shadow` column is present in the schema but not in the feature list, so the LGBM ignores it.

After round-2 has written the new OOF scores back to the sidecar, two FDR variants are logged as a diagnostic before shadow rows are removed.

FDR-A — real-only

Filter `is_shadow == true` out first. Compute the standard target/decoy q-value on the real subset:

$$\text{FDR}_A(s) = \frac{\#\{\text{real decoys with score} \geq s\}}{\#\{\text{real targets with score} \geq s\}}$$

Report the maximum count of real targets seen at any s where $\text{FDR}_A(s) \leq q^*$ (default $q^* = 0.01$).

FDR-B — shadow-inclusive

Do the ratio on **all** rows including shadows on both sides:

$$\text{FDR}_B(s) = \frac{\#\{\text{all decoys with score} \geq s\}}{\#\{\text{all targets with score} \geq s\}}$$

Then count how many **real** targets sit above the score where $\text{FDR}_B \leq q^*$.

FDR-B is the literal answer to “*even though we doubled the decoy count, do more real targets still pass the 1% bar?*” If the round-2 LGBM is well-calibrated on the enlarged set, `shadow_target` rows scoring similarly to real targets can compensate for the `shadow_decoys` in the ratio, and FDR-B can even exceed FDR-A.

Downstream cleanup

After the diagnostics fire, `remove_shadow_decoys!` filters out `is_shadow == true` rows from every fold-file AND from its sidecar (row order must stay aligned). The schema returns to what it was before injection. Chromatogram integration, protein inference, and the output writers see the same data they always did.

Empirical result (FTR-27, MBR off)

Same 27 files, same library, three-species mixture (PA / SAu / UP high, 9 replicates each). Reported: correct IDs (target passes $q \leq 0.01$ with expected-species membership) and cross-species false transfers (target passes but with a wrong-species assignment).

variant	correct	wrong (cross-species)	LGBM gain on mbr_score
develop (no round-2)	136,878	881	—
shadow + knn5_mean	153,393	1,192	12.5%
shadow + global_max	159,644	1,823	13.0%
k=5 mean (no shadow)	168,776	1,283	85.6%

shadow + global_max recovers +**16.6%** correct IDs vs the develop baseline while keeping the LGBM’s reliance on the cross-run feature below 15%. It out-recovers shadow + knn5_mean (+4% more correct) but at ~53% higher cross-species FTR — global_max is a stronger transfer signal AND a stronger leakage signal.

On this dataset the unregularised k=5 mean still wins the raw count contest. The hypothesis under active test is that shadow + global_max will pull ahead on datasets where the top-K nearest neighbors bleed cross-condition — the regularisation protects against precisely the leakage that global_max invites.

Config knobs (all in two_round_scoring.jl)

```
const KNN_SCOPE      = :all_runs      # :nearest_k | :all_runs
const KNN_AGG        = :max           # :mean | :median | :quantile | :max
const KNN_K          = 5              # only when KNN_SCOPE === :nearest_k
const KNN_Q          = 0.75f0        # only when KNN_AGG === :quantile
const SHADOW_DECOY_MODE = :symmetric # :none | :symmetric
```

Column name auto-derives from these to keep round-2 log lines labeled correctly (:global_max, :knn5_mean, :knn10_median, etc.). Setting SHADOW_DECOY_MODE = :none turns off shadow injection while leaving the cross-run feature intact.